

College of Industrial Technology
King Mongkut's University of Technology North Bangkok

Seat No.

Midterm Examination of Semester 1

Year 2025

Subject: 031001121 Chemistry I

Sections 15-17

Date: August 6, 2025

Time 13.00-15.00

Examination Setter Name: 1. Assoc. Prof. Dr. Nathjanan Jongkon

2. Asst. Prof. Dr. Sunanta Chuayprakong

Student Name.....ID.....Class.....No.....

Directions;

1. This examination paper has not been edited. If any errors are identified, students are advised to exercise discretion in addressing them.
2. The examination consists of 8 pages (including the cover page). The total score is 40 points.
3. Complete all questions in this examination paper.
4. Books, documents and lecture notes are NOT allowed.
5. Electronic calculators are NOT allowed.
6. Students must remain in the examination room for no less than 1 hour from the start of the examination.
7. Do not open or start the examination before permission is granted. Students must strictly follow the instructions of the examination.
8. Access to restroom facilities during the examination is not permitted, except in cases of emergency
9. Students are NOT allowed to take the examination paper or copy the examination paper out of the examination room. Otherwise, it will be considered cheating.
10. Usage of all communication devices during the examination is strictly prohibited.
11. Smart watches (electronic watches) are NOT permitted during the exam.

Cheating in examinations is considered a serious offense and
carries the penalty of expulsion from student status.

This exam paper is part of a set used to assess student abilities as follows:

		Exam
CLO 1	Explain principle of chemistry, chemical change and physical change of matter and phenomena.	Questions in Parts 1-2.
CLO 2	Explain classification of matter and selection of proper separation method. Explain energy change of system.	Questions in Parts 1-2.
CLO 3	Explain atomic structure and electron configuration to explain any scientific phenomena.	Questions in Parts 1-2.
CLO 4	Explain and calculate relationship of any matters in chemical reaction to apply in workplace and in scientific industry.	Questions in Parts 1-2.
CLO 5	Apply basic chemistry to other subjects related.	Questions in Parts 1-2.
CLO 6	Realize to have self-discipline, be on time, and be responsible to yourself and society.	Questions in Parts 1-2.

score	Part 1	Part 2	Total

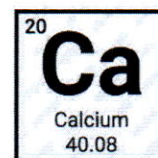
Given: Speed of light; $c = 3.0 \times 10^8$ m/s, Plank's constant; $h = 6.6 \times 10^{-34}$ J.s
Rydberg constant (R_H) = 2.18×10^{-18} J

Part 1. Multiple Choice (5 points)

Choose the letter of the choice in the answer table that best completes the statement or answers the question.

Question no.	a.	b.	c.	d.
1				
2				
3				
4				
5				
6				
7				
8				
9				
10				

1. Which of the following is NOT an example of matter?
a. air b. salad c. heat d. salt water
2. Which statement is an identifying characteristic of a mixture?
a. A mixture can consist of a single element.
b. A mixture can be separated by physical means.
c. A mixture must have a definite composition by weight.
d. A mixture must be homogeneous.
3. Crude oil can be separated by a physical method which is known as _____.
a. filtration. b. crystallization. c. electrolysis. d. distillation.
4. In the plum pudding model, how were the electrons described?
a. Randomly distributed throughout the atom. b. Located in the nucleus.
c. Orbiting the atom in fixed paths. d. Not present in the atom.
5. What is a shape of atomic orbital of helium atom?
a. Sphere b. Dumbbell c. Plum pudding d. Cloverleaf
6. Which of the following is not a physical change?
a. Milk souring. b. Boiling water. c. Cutting wood. d. Broken glass.
7. The formula for manganese (II) sulfate is _____.
a. MnS . b. MgS . c. $MnSO_4$. d. $MgSO_4$.
8. In which change is energy transferred from the surroundings to the substance?
a. Sublimation. b. Condensation. c. Freezing. d. Deposition.
9. What is NOT true about calcium ion?
a. It has 20 protons. b. It has 20 electrons. c. It has 20 neutrals.
d. There are 40 particles at the nucleus of calcium ion.
10. Which of the following atoms has three unpaired electrons?
a. Oxygen (O) b. Boron (B) c. Nitrogen (N) d. Lithium (Li)



Part 2. Read the construction carefully and complete the information required

(35 points)

1. Identify the following as a **chemical** or a **physical** change and fill changes in the right column. (2 points)

1.1 NaCl (Table Salt) dissolves in water.	
1.2 Thomas bakes a cake.	
1.3 Paper towel absorbs water.	
1.4 Kimberly lights fireworks.	

2. Fill the information to the table below. (3 points)

Change of State	Description	Exothermic or endothermic Process
Solid – Liquid		
	Freezing	
	Boiling	
Gas – Liquid		
	Deposition	
Solid – Gas		

3. Use the name and symbol below to give the formulae or the name of the following compounds. (5 points)

Copper (Cu) Barium (Ba) Gold (Au) Iron (Fe)

Sodium hydride _____ AuPO_4 _____

Copper (II) sulfate _____ Fe_2O_3 _____

Strontium phosphate _____ NH_4NO_3 _____

Barium hydroxide _____ SO_2 _____

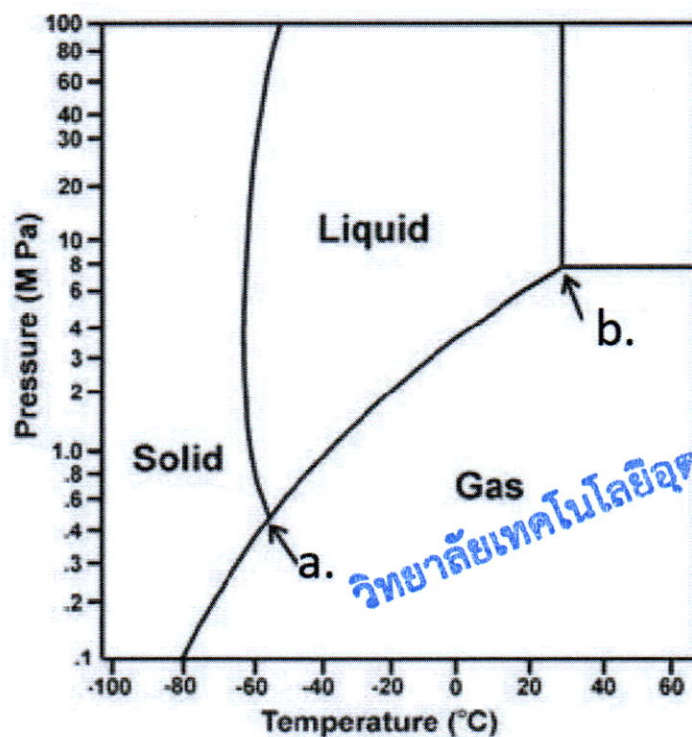
Tetraphosphorus decaoxide _____ N_2O _____

4. Match the best answer and put the answer into the blank. (3 points)

- A. Evaporation A. Fractional distillation B. Chromatography C. Electrolysis
D. Filtration E. Magnetic separation F. Centrifugation G. Crystallization

- _____ 4.1 The best way to get sand from wet sand.
_____ 4.2 Separate iron from iron in soil.
_____ 4.3 What is the separation method involving spinning?
_____ 4.4 Several liquids having different boiling points can be separated by _____.
_____ 4.5 _____ technique can be used to separate mixed food coloring.
_____ 4.6 Electricity must be applied to separate water. This separation method is _____.

5. Answer the questions below in relation to the phase diagram of carbon dioxide.
(3 points)



- 5.1 What the letter "a" represents? _____
5.2 What the letter "b" represents? _____
5.3 What is a normal boiling point? _____
5.4 At what temperature and pressure do all three phases coexist? _____
5.5 At what temperature gas and liquid phases coexist? _____

- 6 Using the Bohr model of hydrogen, what color of light is *given off* or *absorbed* by a hydrogen atom as is electron falls from $n=5$ to $n=2$? (3 points)

($h = 6.6 \times 10^{-34} \text{ J}\cdot\text{s}$, $R_H = 2.18 \times 10^{-18} \text{ J}$ and $c = 3 \times 10^8 \text{ m/s}$)

Color	Wavelength (nm)
red	630–750
orange	590–630
yellow	570–590
green	490–570
blue	450–490
indigo	420–450
violet	380–420

- 7 According to Rutherford's model, draw and specify all fundamental particles of oxygen atom and oxygen ion. (2 points)

oxygen atom	oxygen ion

8. Draw a diagram for carbon (C) atom based on the Plum Pudding model (1 point)

9. Write electron configuration of aluminium atom and specify how many electrons occupy p orbitals? (1 point)

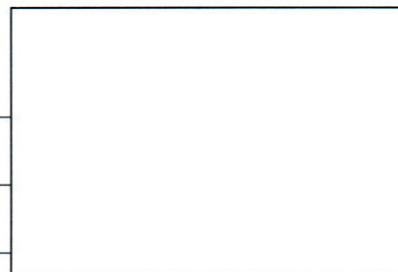
10. Write electron configuration of a potassium atom and specify how many electrons are related to M shell? (1 point)

11. What elements have the following electron configurations? (1 point)

11.1 [He] $2s^2 2p^5$

11.2 [Ne] $3s^2$

12. If principal quantum number (n) is equal to 2, what are all possible angular momentums (l) and how do their orbital shape look? (2 points)



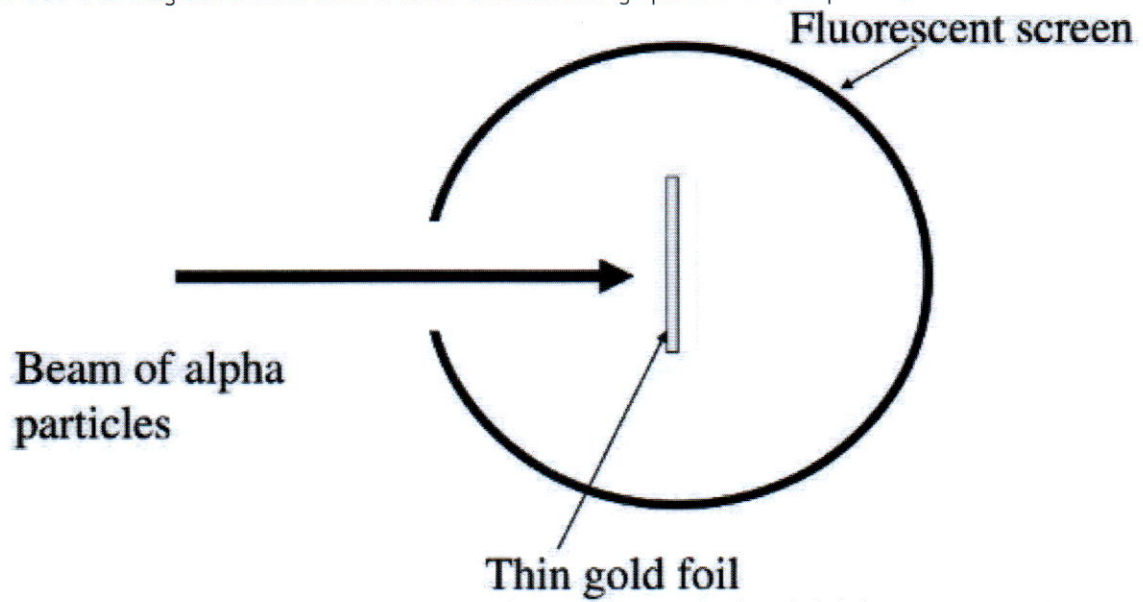
13. Write all possible quantum numbers including n, l, m_l and m_s for Lithium ion. (1 point)

14. Provide 2 characteristics of each following. (3 points)

14.1 Elements in Group IIA	
14.2 Halogen	
14.3 Nobel gases	

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15. Use the diagram below and answer the following questions. (2.5 points)



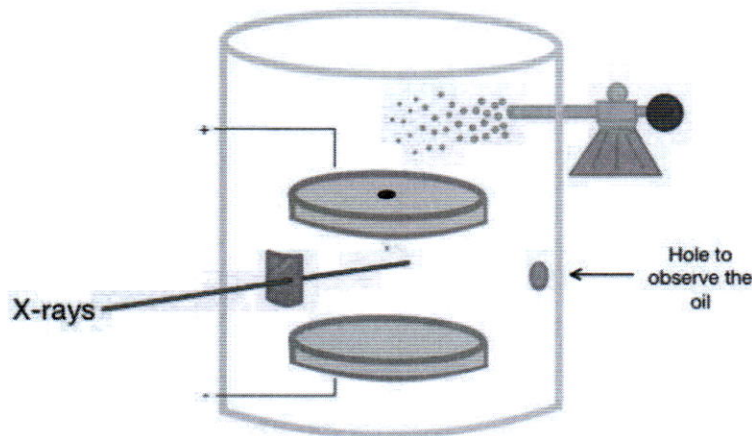
15.1 According the experiment setup above, who initiated and did the experiment?

15.2 What would happen after the beam of alpha particles hit the thin gold foil?

Write or draw the expected result in the diagram above and explain why.

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16. Use the oil drop experiment diagram as shown below and indicate where ionized oil drops will be seen? Provide a reason to support your answer. (1.5 points)



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